Project review summary

Adaptive LLM Security & Resource Risk Gateway

25 October 2026

What was built

A working gateway in front of Qwen2.5-3B/vLLM on an RTX A4000. It extracts lexical, semantic and resource-demand features, uses CUDA/FP16 MiniLM embeddings, reads live KV-cache occupancy, and routes requests to full, output-capped sandbox or quarantine tiers.

What the lab evidence shows

Experiment	Result	Boundary
Nine GPU resource runs	Benign mean-of-run-means: direct **50.24 s**, static **302 ms**, adaptive **298 ms**; all benign completed	Controlled **256 MiB KV pool**, three runs per condition
Resource demand admission	Both gateway modes blocked all 24 demands per condition; protected peak occupancy **0.881%** vs direct **99.56–100%**	No isolated adaptive benefit, default-cache OOM or P99 proof
External short-email test	Adjusted pair: **61.59% source-positive recall**, **76.66% source-negative specificity**, **69.13% balanced accuracy** on 13,989 reserved rows	Automated source labels and heavy selective cleanup; keep candidate offline
Final software check	**135 lab tests passed in 4.609 s**; all four native model artifact hashes matched	Integrity/regression checks, not security efficacy

What we learned

Resource admission protection works for the measured workload. Injection protection is not yet broadly reliable: strong synthetic/training scores did not transfer consistently, and source labels disagreed with task-aware diagnostic review. Reviewer-supplied cases were recorded with provenance and uncertainty; labels were not silently rewritten.

Submission scope and future work

Long-email offline coverage is now tested: 300 emails, 723 windows, 130,735 content-token positions. Max pooling increases positive flags but also flags most source negatives; the live gateway is unchanged. General conversation/trust-boundary protection, GPU cancellation on timeout, independent-source security evaluation and a causal adaptive-threshold advantage remain future work. No zero-total-GPU-cost, measured dollar saving or production-ready security claim is made.

Deliverable: technical report, figures, evidence index and reproducibility logs. Phase 2 content-only transfer: 2,000 emails; adjusted recall 53.92%, specificity 84.26%. Opt-in live extraction improved 24.62 to 18.84 ms; 39 routing parity cases and 138 cloud CPU tests passed. The sub-5-ms target remains unmet.